

Meningoencephalitis (MUE)

What is it?

Meningoencephalitis means inflammation of the brain (*encephal-*) and its protective covering (*meninges*). When the spinal cord is involved it is called **meningomyelitis**. Inflammation can be caused by:

- **Infection** (virus, bacteria, fungus or protozoa), or
- An **auto-immune** reaction — the immune system overreacts and attacks the brain. This is called **MUE** (meningoencephalitis of unknown etiology).

Signs

Signs depend on which part of the brain is inflamed, and several areas can be affected at once.

- **Front of the brain:** seizures, confusion, drowsiness, behavior change, pain
- **Back of the brain:** poor balance, weakness, abnormal eye movements, pain

Diagnosis

- **MRI** of the brain (or spine) to find inflammation and rule out tumors or fluid buildup
- **Spinal tap** to analyze spinal fluid
- **Testing for common infectious diseases**

Steroids given before testing (prednisone, prednisolone, dexamethasone) can make spinal fluid results less accurate.

Treatment

- **Infectious:** antibiotics or antifungal drugs.
- **Auto-immune (MUE):** medications that calm the immune system. We start with a higher steroid dose to

"put out the fire." Because long-term high-dose steroids cause side effects, we usually add a second (sometimes third) immunosuppressive drug to keep the disease controlled while slowly weaning the steroid.

What to expect — "the rule of thirds"

For auto-immune meningoencephalitis:

- About ⅓ do not improve and worsen within days
- About ⅓ improve but have relapses
- About ⅓ recover well with ongoing medication and care

Vaccinations

- **Do not vaccinate while your pet is being treated** for auto-immune ME.
- Vaccines may trigger a relapse, which can be more serious and harder to treat than the first episode. We recommend only the **most necessary vaccines**, decided case by case.
- **Antibody titer** blood tests can show whether your pet is still protected against distemper, parvovirus and adenovirus. Titers cannot replace the **legally required rabies** vaccine.
- Indoor pets have lower infection risk than active, outdoor pets. We cannot predict whether a vaccine will cause a relapse — there are risks either way, and we will help you decide.

Call us if

Seizures, worsening balance, new weakness, not eating, or possible medication side effects.